

From Output-Only Transmissibilities to a Finite-Record Estimator

Saturday 26 September 2026: estimator construction, twin validation and Gate G2

Technical writeup, continued

This note continues the Friday writeup. Friday established what information an ideal, noise-free transmissibility carries about (β, R) . Saturday asked whether that transmissibility can actually be *estimated* from finite, noisy, stochastic multi-sensor records, and what its real bias and covariance are. Four files were built and validated in MATLAB (S1 to S4), and the Gate G2 Monte Carlo experiment was run. The main new result is that the estimator's long-record bias is a deterministic, accurately predicted spectral-window effect, which fixes the production data representation for Sunday. Colour convention as before: findings in green, caveats and scope limits in red, definitions and theory in blue.

1 Why the pipeline pivoted from SSI poles to transmissibilities

The project began with Stochastic Subspace Identification (SSI): fit output-only system poles to the IMU records and invert the poles for the floe parameters. Friday's evidence removed poles from the critical path:

1. **Nothing to fit.** At p_0 the sensor acceleration FRFs have no isolated resonance: no peak of $\|\mathbf{H}\|$ exceeds 0.3 dB prominence, and the one broad pitch-like feature has a half-power width of 3.43 rad/s. Local pole fitting per resonance window had no windows.
2. **The information is in the spatial ratios.** Under a single common incident field every output-only statistic is a function of the transmissibilities $T_j = H_j/H_r$ and the reference log-level $\ell = \ln(|H_r|^2 S_u)$. With an unknown but smooth sea spectrum S_u , ℓ adds nothing (Friday file 5d, rows D2 to E), so *any* output-only method, SSI included, is bounded by the transmissibility information: $\sigma_{\ln \beta} = 3.985$, $\sigma_{\ln R} = 0.784$ per unit relative noise density, versus 2.315 and 0.263 for the complex FRF reference.
3. **Coloured seas are a trap for poles.** Output-only poles are only system poles if the excitation is broadband enough; a JONSWAP-like sea can put its own colour into the fitted poles.
4. **The hardware is a multi-sensor array**, so the spatial transmissibilities are directly measurable.

Two decisions carried over from Friday: γ is fixed from measured thickness, so $\theta = [\ln \beta, \ln R]^T$; and SSI is retained only as an optional comparison (do poles track the floe or the sea under coloured forcing?).

Transmissibility. With reference sensor r (s26) and the single-common-input model $y_j = H_j u$,

$$T_j(\omega) = \frac{H_j(\omega)}{H_r(\omega)} = \frac{S_{jr}(\omega)}{S_{rr}(\omega)}, \quad S_{jr} = E[Y_j Y_r^*] = H_j H_r^* S_u, \quad S_{rr} = |H_r|^2 S_u, \quad (1)$$

so the unknown wave spectrum S_u cancels. H_j is the acceleration FRF (m s^{-2} per metre of incident amplitude) under the $e^{+i\omega t}$ convention of the EMM. The ratio discards the common factor H_r : absolute level and phase relative to the incident wave.

2 The Saturday pipeline

Two independent paths meet in the inversion:

$$\underbrace{(\beta, R) \xrightarrow{\text{EMM}} T_{\text{model}}(\omega_k)}_{\text{S1: model side}} \quad \text{and} \quad \underbrace{(\beta, \gamma, R, S_u, \text{noise}) \xrightarrow{\text{S3}} Y(t) \xrightarrow{\text{S2}} \hat{T}(\omega_k)}_{\text{data side}}, \quad (2)$$

and S4 measures, by Monte Carlo over S3 records, the bias, covariance, coherence and record-length dependence of $\hat{T}$. Three frequency grids are deliberately kept apart: the EMM *node* grid (where the EMM is solved, cached), the *synthesis* grid (FFT bins of a record, H interpolated from the nodes) and the *Welch* grid (set by S2, on which S1 is evaluated directly, so no interpolation ever enters the inversion).

2.1 S2: Welch transmissibility estimator (extractTransmissibility.m)

Estimator. Records of N samples at Δt are split into segments of L samples with fractional overlap (here 50%), each segment mean-removed and multiplied by a periodic Hann window $w_n =$

$\frac{1}{2} - \frac{1}{2} \cos(2\pi n/L)$. With segment DFTs $X_{j,s}(\omega_k)$ at $\omega_k = 2\pi k/(L\Delta t)$,

$$\hat{S}_{jr}(\omega_k) = \frac{2\Delta t}{2\pi U} \frac{1}{K} \sum_{s=1}^K X_{j,s} X_{r,s}^*, \quad U = \sum_n w_n^2 = \frac{3}{8}L, \quad \hat{T}_j = \frac{\hat{S}_{jr}}{\hat{S}_{rr}}. \quad (3)$$

The orientation $E[Y_j Y_r^*]$ returns H_j/H_r ; the conjugate orientation would return its conjugate (phase error π in the delay test). Spectra are one-sided per rad/s, so white noise of standard deviation σ_n has

$$S_{nn} = \frac{\sigma_n^2 \Delta t}{\pi}. \quad (4)$$

Reference-noise correction. Independent sensor noise does not bias $\hat{S}_{jr}$ but inflates $\hat{S}_{rr}$ by S_{nn} , attenuating $\hat{T}$ by $S_{rr}/(S_{rr} + S_{nn})$. With a known noise PSD, $\hat{T}_{j,\text{corr}} = \hat{S}_{jr}/(\hat{S}_{rr} - S_{nn})$; bins with $\hat{S}_{rr} - S_{nn} < 0.1 \hat{S}_{rr}$ are flagged invalid and set to NaN, never used silently.

Diagnostics. Coherence $\gamma_{jr}^2 = |\hat{S}_{jr}|^2/(\hat{S}_{jj}\hat{S}_{rr})$; the effective number of averages for overlapped windows (Welch 1967) $N_{\text{eff}} = K/(1 + 2 \sum_{l \geq 1} (1 - l/K) c_l^2)$ with $c_l = \sum_n w_n w_{n+lh}/U$ and hop h ; and the Bendat–Piersol diagonal variance

$$E|\hat{T}_j - T_j|^2 \approx |T_j|^2 \frac{1 - \gamma_{jr}^2}{\gamma_{jr}^2 N_{\text{eff}}}, \quad (5)$$

a diagnostic only: it ignores the correlation between the T_j that share the reference channel.

2.2 S1: model-side feature map (transmissibilityFeatureVector.m)

Parameters. $p = P_{\text{base}}$, $p_{\text{idx}} = e^\theta$ with default $\text{idx} = [1, 3]$, so $\theta = [\ln \beta, \ln R]^T$ and γ comes from P_{base} ; $\text{idx} = 1:3$ restores the three-parameter map, so fixing γ stays reversible.

Frequency-major stacking (frozen). Per frequency,

$$z_k = [\text{Re } T_1, \text{Re } T_2, \text{Re } T_3, \text{Im } T_1, \text{Im } T_2, \text{Im } T_3]_{\omega_k}^T \in \mathbb{R}^6, \quad f_T = [z_1^T, \dots, z_{n_F}^T]^T. \quad (6)$$

Each z_k is exactly the variable whose 6×6 covariance Σ_k S4 estimates, so $\Sigma_T = \text{blkdiag}(\Sigma_1, \dots, \Sigma_{n_F})$ needs no index permutation. A NaN entry marks both its real and imaginary parts.

2.3 S3: stochastic EMM twin (synthesiseTwinRecords.m)

Node FRF and interpolation. The EMM is solved on nodes 0.1 rad/s apart spanning the validated band $[2.978, 8.501]$ rad/s (57 solves, cached on disk). The cache key is p , sensors, band, node spacing, EMM options and interpolation-check settings; a mismatch forces a new solve. The smoother $H_d = H_a/(-\omega^2)$ is spline-interpolated in real and imaginary parts and $H_a = -\omega^2 H_d$ restored.

Record model. On the record's FFT grid,

$$y_j(t) = \text{Re} \sum_k w_k \sqrt{S_u(\omega_k) \Delta \omega} H_j(\omega_k) \xi_k e^{i\omega_k t} + n_j(t), \quad (7)$$

with one complex Gaussian ξ_k *shared* by all sensors (single input, rank-one cross-spectrum), S_u white or JONSWAP (normalised to $H_s = 4\sqrt{m_0}$ over the full spectrum), an amplitude taper w_k that ramps to zero over 0.2 rad/s *inside* the band edges so that no unvalidated EMM frequency is excited, and independent white sensor noise n_j . Wave and noise use separate random streams: for a given seed the wave realisation is identical at every noise level (common random numbers).

File	MATLAB validation (all tester groups pass)
S2 estimator (11 groups)	Pure-delay phase error 8.8×10^{-3} rad; known modal system median error 4.8×10^{-3} , max 5.8×10^{-2} ; PSD normalisation 0.997 of $\sigma^2 \Delta t / \pi$; non-reference noise: no bias; reference-noise attenuation 0.648 observed vs 0.655 predicted; correction 34.8% $\rightarrow$ 2.8%; coloured input: same mean, variance $\times 8.6$ in low-energy bins; BP/MC variance 0.98
S1 feature map (7 groups)	Agreement with H_j/H_r from <code>computeSensorFRF</code> : 4.3×10^{-16} ; reference change $T_j^{(a)} = T_j^{(r)}/T_q^{(r)}$: 7.5×10^{-17} ; saved Friday 5b data: 6.3×10^{-16}
S3 twin (7 groups)	FFT-bin ratio vs H_j/H_r : 2.4×10^{-13} ; ensemble variance ratios [1.02, 1.01, 1.01, 1.02]; S3 $\rightarrow$ S2 Welch T error median 2.2×10^{-3} ; interpolation error at 0.25 rad/s nodes $\epsilon_T = 2.5 \times 10^{-5}$; node T vs S1 1.3×10^{-15} ; cache 8.0 $\rightarrow$ 0.16 s; stale cache keys rejected
Production twin (in S4)	Interpolation check $\epsilon_T = 2.8 \times 10^{-6}$; S1 EMM features vs interpolated truth on the Welch grid: max 2.0×10^{-6}

Table 1: Validation of the three building blocks. Relative errors unless stated.

S1, S2 and S3 validated and frozen. Model and data features share one tested frequency-major convention on one grid; the twin follows the same transmissibility map as S1 to 2×10^{-6} , far below any stochastic error studied below.

3 Gate G2 experiment (S4, runTransmissibilityEstimatorTest.m)

Parameter point, sensors	$p_0 = (4.6985 \times 10^{-5}, 1.4548 \times 10^{-3}, 0.3830)$; Level 2C [2 6 14 26], reference s26
Welch	$\Delta t = 0.1$ s, $L = 512$, 50% overlap, band [3, 8.5] rad/s: 45 bins, 3.068 to 8.468
Record lengths, repeats	$N = 4096, 16384, 65536$ (410 s, 27 min, 109 min), $N_{MC} = 200$ each
Inputs	JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s, $\gamma_J = 3.3$; white matched in in-band variance
Sensor noise	$\sigma_n = 6.67 \times 10^{-3}$ m/s ² on every channel (10% of JONSWAP s26 std), identical for white and JONSWAP
Reference SNR G_r/S_{nn}	white 2.8 to 27.9 dB; JONSWAP -3.5 to 33.4 dB
Cases	W-clean, J-clean, W-noise, W-noise-c, J-noise, J-noise-c (c = corrected, same records)
Segment sweep	$L = 256, 512, 1024$ at $N = 65536$ for W-clean, J-clean, J-noise

Table 2: G2 configuration.

The 10% noise level is a controlled stress test chosen to make the noise-limited regime visible, not an IMU specification. At 2% (dry run) noise barely mattered.

Statistics per bin (valid realisations only). With $z_k^{(i)}$ the draws and z_k the truth: bias $b_k = \bar{z}_k - z_k$, reported relative as $\|b_k\|/\|z_k\|$; covariance Σ_k and relative standard deviation $\sqrt{\text{tr } \Sigma_k}/\|z_k\|$ (for *one* record); Hotelling $T_k^2 = n b_k^T \Sigma_k^{-1} b_k$ compared with 16.81, the asymptotic χ_6^2 99% point (a diagnostic, not an exact finite-sample test); inter-frequency correlation $\rho_l = \text{mean}_i |\text{corr}(z_{k,i}, z_{k+l,i})|$, whose Monte Carlo floor for independent bins is $\sqrt{2/(\pi N_{MC})} = 0.06$.

4 Central result: Welch measures a spectral-window average

Welch-expected transmissibility. A Hann segment does not sample T at ω_k ; it averages the cross- and auto-spectra across the window's spectral kernel. For a record synthesised on the fine grid ω_q with tapered input PSD $G_q = S_u(\omega_q)w_q^2$, the expected segment spectra give

$$T_{\text{pred},j}(\omega_k) = \frac{\sum_q K(\omega_k - \omega_q) G_q H_j(\omega_q) H_r^*(\omega_q)}{\sum_q K(\omega_k - \omega_q) G_q |H_r(\omega_q)|^2 + S_{nn}}, \quad K(\Delta\omega) = \frac{|W(\Delta\omega \Delta t)|^2}{N U}, \quad (8)$$

where $W(\nu) = \frac{1}{2}D(\nu) - \frac{1}{4}D(\nu - \frac{2\pi}{L}) - \frac{1}{4}D(\nu + \frac{2\pi}{L})$, $D(x) = (1 - e^{-ixL})/(1 - e^{-ix})$, is the DTFT of the periodic Hann window and $\sum_{d=0}^{N-1} |W(2\pi d/N)|^2 = NU$ normalises K (both verified numerically in the script). S_{nn} appears only in uncorrected noisy cases. Eq. (8) has no free parameters and separates

- **Welch-resolution bias:** H_j/H_r and the input weighting $G_q|H_r|^2$ vary across the kernel main lobe (about ± 2 bins), so the ratio of averages is not the point ratio;
- **reference-noise bias:** the S_{nn} term in the denominator;

from the residual $\bar{z}_k - z_{\text{pred},k}$, the finite-sample ratio bias, which falls with N_{eff} .

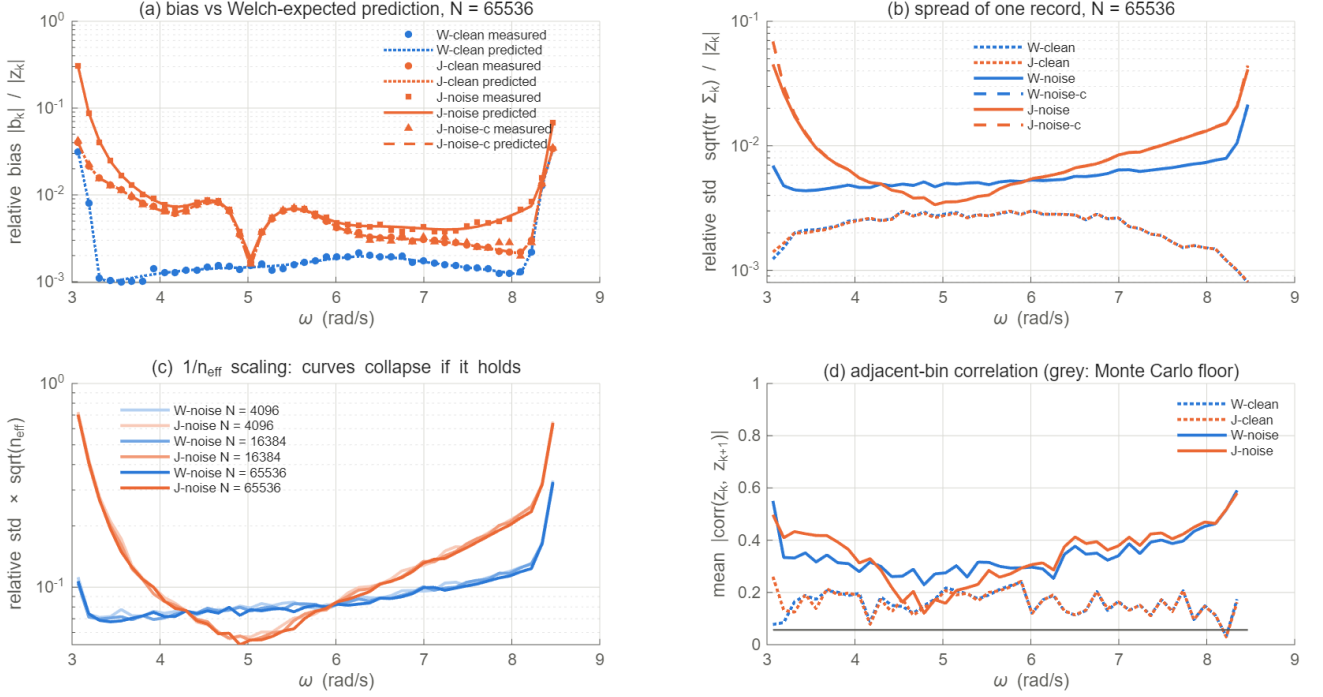


Figure 1: G2 at $L = 512$. (a) Relative bias at $N = 65536$: markers measured, lines Eq. (8). (b) Relative std of one record. (c) Relative std $\times \sqrt{N_{\text{eff}}}$ for the three record lengths (collapse means $\text{Var} \propto 1/N_{\text{eff}}$). (d) Adjacent-bin correlation with the Monte Carlo floor (grey).

Case ($N = 65536$)	rel. bias	residual	rel. std	bias/std	sig. bins: $z \rightarrow z_{\text{pred}}$	BP/MC
W-clean	1.53×10^{-3}	1.2×10^{-4}	2.54×10^{-3}	0.64	1.00 $\rightarrow$ 0.07	1.30
J-clean	5.00×10^{-3}	1.3×10^{-4}	2.55×10^{-3}	1.93	1.00 $\rightarrow$ 0.04	1.30
W-noise	2.94×10^{-3}	3.3×10^{-4}	5.23×10^{-3}	0.60	1.00 $\rightarrow$ 0.00	1.06
W-noise-c	1.59×10^{-3}	3.3×10^{-4}	5.23×10^{-3}	0.29	1.00 $\rightarrow$ 0.00	1.06
J-noise	6.09×10^{-3}	4.5×10^{-4}	6.88×10^{-3}	1.09	1.00 $\rightarrow$ 0.00	1.06
J-noise-c	5.16×10^{-3}	4.6×10^{-4}	6.91×10^{-3}	0.76	1.00 $\rightarrow$ 0.00	1.04

Table 3: Medians over the 45 bins at the longest record. “Residual” is $\|\bar{z} - z_{\text{pred}}\|/\|z\|$; “sig. bins” is the fraction of bins with $T^2 > 16.81$ against the point truth and against Eq. (8). All corrected estimates were valid (fraction 1.000).

The long-record bias is not an estimator failure. Against the point ratio H_j/H_r every bin is significantly biased at $N = 65536$; against Eq. (8) only 0 to 7% are, and the residual is 5 to 40 times smaller than the bias. At the lowest JONSWAP bin (3.068 rad/s, -2.2 dB) the uncorrected bias is 0.304, predicted 0.304. Because the kernel width is set by L , not N , a longer record shrinks the scatter but not the bias (Table 4).

J-clean, $L = 512$	$N = 4096$	$N = 16384$	$N = 65536$
median relative bias	4.83×10^{-3}	5.18×10^{-3}	5.00×10^{-3}
median relative std (one record)	1.05×10^{-2}	5.02×10^{-3}	2.55×10^{-3}

Table 4: Bias is fixed by the segment length; scatter falls as $N_{\text{eff}}^{-1/2}$.

5 Variance, colour and noise correction

Variance behaves conventionally (G2.2, G2.4). The log-log slope of median $\text{tr} \Sigma_k / \|z_k\|^2$ against N_{eff} is -1.00 to -1.02 in all six cases, and $\sigma_{\text{rel}} \sqrt{N_{\text{eff}}}$ is constant across record lengths (Fig. 1c). The Bendat–Piersol diagonal matches Monte Carlo to 1.06 in both noisy cases. Even noise-free records have finite variance: the random input weights the frequencies inside each kernel differently in every segment (clean BP/MC 1.30).

Colour does not change T , it changes where $\hat{T}$ is useful (G2.5). In 99 noise-dominated (bin, channel) pairs the slope of $\log(\text{Var}_J/\text{Var}_W)$ against $\log(G_{rJ}/G_{rW})$ is -0.82 (theory -1 for $\text{Var} \sim S_{nn}/G_r$). JONSWAP reference SNR spans -3.5 to 33.4 dB (Fig. 2a), so its low-frequency and edge bins are noise-limited, while near the peak it beats white.

Reference-noise correction works (G2.3). In the bins with predicted noise bias above 1% (2 white, 7 JONSWAP), the noise part of the bias, $\|\bar{z} - z_{\text{pred, clean}}\|/\|z\|$, falls from 1.21×10^{-2} to 6.0×10^{-4} (white) and from 3.59×10^{-2} to 1.05×10^{-3} (JONSWAP), factors of 20 and 34. At 3.068 rad/s the correction takes the bias from 0.304 to 0.0415, leaving the clean resolution bias of 0.0402. No corrected bin was invalidated, even at -3.5 dB (Fig. 2b).

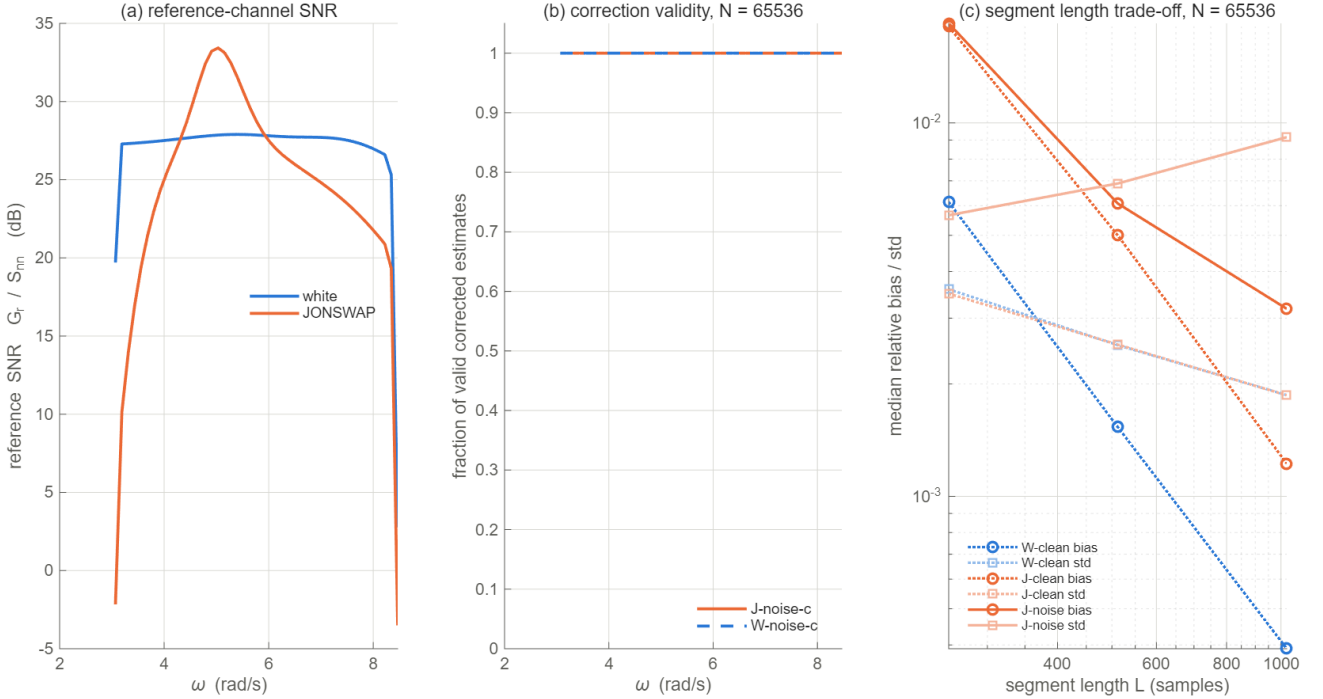


Figure 2: (a) Reference SNR G_r/S_{nn} . (b) Fraction of valid corrected estimates at $N = 65536$. (c) Segment-length trade-off at $N = 65536$: median relative bias (circles) and one-record std (squares).

6 Segment length and inter-frequency correlation

L	bins	N_{eff}	W-clean			J-clean			J-noise		
			bias	pred	std	bias	pred	std	bias	pred	std
256	22	484	6.14	6.10	3.59	18.1	18.2	3.49	18.5	18.4	5.66
512	45	242	1.53	1.51	2.54	5.00	5.01	2.55	6.09	6.02	6.88
1024	90	120	0.39	0.37	1.87	1.22	1.21	1.87	3.18	3.14	9.16

Table 5: Segment-length sweep at $N = 65536$: medians over bins, all $\times 10^{-3}$ (relative). “pred” is Eq. (8); J-noise is uncorrected.

$L = 1024$ is the production segment length. Doubling L halves the kernel width and cuts the resolution bias by about 4 (roughly L^{-2}), exactly as predicted. In the noise-dominated case the scatter rises as $\sqrt{L}$ (fewer averages); in the clean cases it *falls*, because less spectral mixing outweighs the lost averages. At $L = 1024$ the JONSWAP resolution bias is 1.2×10^{-3} , below the noisy one-record scatter 9.2×10^{-3} .

L	Case	Full grid			Every second bin	
		ρ_1	ρ_2	ρ_3	ρ_1	ρ_2
512	J-clean	0.16	0.15	0.06	0.14	0.05
512	J-noise	0.38	0.06	0.06	0.07	0.05
1024	J-clean	0.17	0.13	0.05	0.13	0.06
1024	J-noise	0.42	0.06	0.06	0.06	0.06

Table 6: Median inter-frequency correlation (`checkSegmentSweepCorrelation.m`, $N = 65536$, $N_{\text{MC}} = 200$, floor 0.06). The decimated lag 1 is the full-grid lag 2.

Adjacent bins are correlated; every second bin is not. On the full noisy grid $\rho_1 = 0.42$ at $L = 1024$ (the Hann kernels of neighbouring bins overlap) while ρ_2 is at the floor. Keeping every second $L = 1024$ bin gives $\rho_1 = 0.06$, the Monte Carlo floor. The clean cases keep a weak lag-2 correlation (0.13) from the shared input weighting, but the inversion scenario is the noisy one.

Two cautions. (i) Treating every full-grid bin as independent would overstate the information; decimation removes the correlation but also discards data, and how much must be quantified in U1 against a covariance that keeps the adjacent-bin coupling. (ii) The largest biases, 4.0% at 3.07 rad/s and 3.4% at 8.47 rad/s, sit inside the synthesis taper ramps [2.978, 3.178] and [8.301, 8.501] rad/s. The ramps are a twin device, not physics, so the inversion band must exclude them: 3 of the 45 decimated $L = 1024$ bins.

7 Gate G2 outcome

Check	Result ($N = 65536$)	
G2.1 bias model	significant-bias fraction $1.00 \rightarrow 0.00$ –0.07 against Eq. (8)	PASS
G2.2 $\text{Var} \propto 1/N_{\text{eff}}$	slopes -1.00 to -1.02 , all six cases	PASS
G2.3 noise correction	noise part of bias reduced $20\times$ (W), $34\times$ (J)	PASS
G2.4 Bendat–Piersol	BP/MC 1.06 (W and J noisy)	PASS
G2.5 colour	slope -0.82 vs -1 , 99 noise-dominated pairs	PASS
G2.6 inter-frequency	$\rho_1 = 0.32$ (W), 0.38 (J) at $L = 512$; floor 0.06	INFO
G2.7 segment length	Table 5	INFO
G2.8 white-weighted smoothing	if the model is kernel-smoothed assuming white S_u but the sea is JON-SWAP, bias median 3.2×10^{-3} remains (raw 5.0×10^{-3})	INFO

Table 7: Gate G2 checks (MATLAB run `transmissibilityEstimatorTest_20260926_135344`).

Gate G2 passed. Complex transmissibilities can be estimated from finite stochastic multi-sensor records under a single common incident field. Their variance scales as $1/N_{\text{eff}}$ and agrees with the Bendat–Piersol diagnostic; spectral colour redistributes that variance through the local SNR; reference noise produces the predicted denominator bias, largely removed when its PSD is known. The remaining discrepancy from the point ratio is a deterministic Welch spectral-resolution effect, accurately predicted by Eq. (8) and strongly reduced at $L = 1024$. G2.8 shows that smoothing the model with an assumed white spectrum does not remove it, so the model side stays the point ratio and the bias is controlled through L .

What G2 did not show. One coherent incident field only (no directional spreading or multiple independent forcing); white, independent sensor noise with a known PSD; 10% noise is a stress case, not hardware validation; everything is local to p_0 , this sensor layout and the validated EMM band; and the per-bin bias versus per-bin scatter comparison says nothing yet about *parameter* bias (Section 9).

8 Production configuration and files

Parameters	$\theta = [\ln \beta, \ln R]^T$, γ fixed provisionally; reference sensor s26
Welch	$\Delta t = 0.1$ s, $L = 1024$, 50% overlap, raw spacing 0.0614 rad/s
Frequencies	every second bin, spacing 0.1227 rad/s, 3.252 to 8.284 rad/s: $n_F = 42$
Features	6 per frequency, $\dim f_T = 252$; noise correction on
Model observable	point ratio $T_j = H_j/H_r$ (S1)
Covariance	production: blkdiag(Σ_k), 6×6 blocks; benchmark: full grid with lag-1 coupling

Table 8: Production configuration after Saturday.

File (folder)	Role	Outcome
extractTransmissibility.m (Transmissibility/) + tester	Welch estimator, noise correction, diagnostics	11 groups pass
stackTransmissibility.m, unstackT ransmissibility.m (Transmissibility/)	Frozen frequency-major layout	tested in S1
transmissibilityFeatureVector.m (InverseMapping/) + tester	Model features $T(\theta)$	7 groups pass
synthesiseTwinRecords.m (Transmissibility/) + tester	Cached stochastic EMM twin	7 groups pass
runTransmissibilityEstimatorTest.m (.../SSITwin/)	G2 Monte Carlo (S4)	5 checks pass
checkSegmentSweepCorrelation.m (.../SSITwin/)	Lag correlation of the L sweep	Table 6

Table 9: Saturday’s files. .../SSI Twin/ is Inverse Mapping/Parameter Inversion/SSI Twin/ (to be renamed Output-Only Twin/); results and the EMM cache are in Inverse Mapping/Parameter Inversion/Results/.

9 Sunday 27 September 2026: U1, U2 and U3 (Gate G3)

U1: covariance-weighted information (runTransmissibilityCovariance.m).

1. **Fresh draws at the production configuration:** 200 records, $N = 65536$, JONSWAP, 10% noise, correction on, $L = 1024$, full 90-bin grid kept (about 2 min, node FRF cached). This removes the proxies of Saturday (the sweep had no corrected case) and gives Σ_k , $C_{k,k+1}$ and the true remaining bias b_k in one pass. The $L = 512$ benchmark reuses the saved S4 draws.
2. **Jacobian** $J_T = \partial f_T / \partial \theta$ by central differences through S1 on the production grid (and the full 90-bin grid for the benchmark), cached; not interpolated from Friday’s 5b data.
3. **Information** $F = J_T^T \Sigma_T^{-1} J_T$ under three treatments: (i) $L = 512$, all bins, block diagonal (optimistic); (ii) $L = 1024$, every second bin, block diagonal (production); (iii) $L = 1024$, all bins, block

tridiagonal

$$\Sigma_{\text{band}} = \begin{bmatrix} \Sigma_1 & C_{12} & & \\ C_{12}^T & \Sigma_2 & C_{23} & \\ & C_{23}^T & \Sigma_3 & \ddots \\ & & \ddots & \ddots \end{bmatrix}, \quad (9)$$

each $C_{k,k+1}$ from the 12×12 pair covariance; if Cholesky fails, shrink towards the block diagonal. Compare $\sigma_{\ln \beta}$, $\sigma_{\ln R}$: if (ii) is within about 20% of (iii), decimation is confirmed; also compare with Friday's surrogate result (5c).

4. **Parameter bias.** Per-bin bias against per-bin scatter is the wrong test: scatter averages down across 42 bins, a systematic bias does not. Project it:

$$\delta\theta_{\text{bias}} = F^{-1} J_T^T \Sigma_T^{-1} b, \quad \text{compare with } \sqrt{\text{diag } F^{-1}}, \quad d_{\text{bias}} = \sqrt{\delta\theta_{\text{bias}}^T F \delta\theta_{\text{bias}}}. \quad (10)$$

$d_{\text{bias}} \ll 1$ keeps the point-ratio model; $d_{\text{bias}} \gtrsim 1$ means the resolution bias must enter the observation model (Scenario W, with a measured S_u).

Σ is specific to p_0 , this sea state and noise level: suitable for U1 and local inversions; for distant truth points the Bendat–Piersol diagonal is the portable fallback.

U2: twin inversion (`runTransmissibilityTwinInversion.m`). A weighted residual wrapper (without modifying `solveInverse.m`); two or three nonlinear inversions from finite records at truth points away from p_0 , then a large linearised Monte Carlo. Gate G3: *do finite-record output-only transmissibilities recover (β, R) with errors consistent with the U1 covariance?*

U3 (stretch): spectrum prior (`runSpectrumPriorInformation.m`). Scenario P: JONSWAP parameters as nuisances with a prior, $F_{\text{eff}} = F_{\theta\theta} - F_{\theta\eta}(F_{\eta\eta} + \Sigma_\eta^{-1})^{-1}F_{\eta\theta}$.

Monday supervisor points. Confirm fixing γ from measured thickness; confirm SSI as an optional comparison only; wave probe in the tank (Scenario W); realistic IMU noise level for the stress test.